# WOCCOM 2026

Workshop and Conference on Computational Mathematics and Modelling

School of Mathematics, Statistics and Computer Science,
University of KwaZulu-Natal, Pietermaritzburg, South Africa.

29 June 2026 — 03 July 2026

**Local Organising Committee:**
Precious Sibanda, Sandile Motsa, Sicelo Goqo, Olumuyiwa Otegbeye, Shina Daniel Oloniiju,
Vusi Magagula, Hloniphile Sithole-Mthethwa, Thama Duba, Yusuf Olatunji Tijani, & Titilayo Agbaje

*The organising committee acknowledges generous sponsorship from the*
*DSI-NRF Centre of Excellence in Mathematical and Statistical Sciences (CoE-MaSS)*

# 1 WELCOME FROM THE CHAIR OF THE LOCAL ORGANIZING COMMITTEE

It is my singular honour and privilege to welcome you all to the *18th Annual Workshop and Conference on Computational Mathematics and Modelling* (WOCCOM 2026), hosted at the Pietermaritzburg Campus of the University of KwaZulu-Natal.

This year marks another milestone for our community. What began in 2008 as a focused initiative to foster growth in computational mathematics among mostly postgraduate students has matured over nearly two decades into a premier annual gathering. Historically, our event has drawn 60-80 participants. This year, for the very first time, our registrations have topped 100, reflecting the vibrant and sustained relevance of this series.

WOCCOM has always been, at its heart, an incubator for the next generation of mathematical talent. We are thrilled to welcome a diverse cohort of postgraduate students and young academics representing institutions across South Africa, the broader SADC region, and the African continent.
Our core mission remains two-fold:

- To teach and equip. The intensive workshop component (29 June -1 July) is specifically designed to introduce innovative numerical methods and computational ideas.
- To connect and collaborate. We intentionally create conditions that foster networking. The true measure of WOCCOM's success is that over the years, these informal interactions have crystallized into formal and informal research collaborations, resulting in a rich portfolio of published research.

The second half of our gathering shifts to the conference component (2-3 July). This year's themes spanning climate models, block hybrid methods for boundary value problems, and the effective use of Large Language Models (LLMs) in numerical approximations are designed introduce and stimulate research in areas we have previously not tackled but are highly relevant in addressing some of the current computational challenges. We extend our sincere gratitude to all the researchers and students who submitted abstracts. We eagerly anticipate the presentations, discussions, and intellectual debates that will undoubtedly unfold over the next few days.

An event of this scale is only possible through collective dedication. I would like to extend my deepest appreciation to workshop facilitators, Dr. Nosipho Zwane, Dr. Shina Oloniiju and Professor Sandile Motsa, for leading the foundational sessions. We are equally indebted to our plenary speakers, Prof. Thando Ndarana (University of Pretoria), Prof. Montaz Ali (University of the Witwatersrand), and Dr. Olumuyiwa Otegbeye (University of the Witwatersrand) who have travelled to share their expertise with us.

Finally, I want to thank the members of the Local Organizing Committee and our sponsors, including the DSI-NRF Centre of Excellence in Mathematical and Statistical Sciences (CoE-MaSS), for their tireless work behind the scenes.

To all our participants, I encourage you to engage fully, ask difficult questions, step out of your comfort zones to forge new networks, and enjoy the beautiful winter warmth of Pietermaritzburg.

As our theme reminds us, *Model today. Understand tomorrow. Act for a sustainable future.*

Welcome to WOCCOM 2026!

**Prof. Precious Sibanda**
Chair: Local Organizing Committee (WOCCOM 2026)
*School of Agriculture and Science*
*University of KwaZulu-Natal*

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## 2 General Information

- **Presentation details:** Please ensure your presentation is loaded onto the available machine before the session in which you are speaking. If you intend to use your own laptop, please check the connections in advance. A laser pointer and clicker will be available. Contributed talks will be kept strictly to 20 minutes (15 minutes for the talk + 5 minutes for questions).
- **Registration:** Registration will take place inside MSB G24 Lab.
- **Talks:** Talks will be held in council chambers and DSLT.
- **Tea & Coffee:** Tea and coffee will be available at the times indicated in the schedule in MSB F15.
- **Lunch Venue:** Lunch will be served in MSB F15.
- **Dinner:** The conference dinner will be held at Golden Horse Casino - Rockafellas from 18:00 – 22:00 on Thursday, 02 July 2026.
  Please note our dress code: Smart attire, no men's sandals, no flip flops, no tracksuit pants or sweatpants, no shorts/mini skirts.
- **Wifi details:** Connect to the `MSCS-AD SSID` network using the following credentials:
  **Username:** `workshop`
  **Password:** `WOCCOM2026`
  Users that have Eduroam access may also be able to use the Eduroam SSID.

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## 3 Workshop

The pre-conference workshop on Generative AI for Scientific Code Generation, climate modelling and hybrid block method to solution of differential equation will take place in **G24 Lab** from Monday, 29 June to Wednesday, 1 July 2026. Sessions will run from 09:00 to 16:30 on all days.

- **Facilitator:** Prof. Sandile Motsa (University of Eswatini).
- **Theme:** The workshop explores the use of large-language model (LLM) tools for developing scripts based on the application of well-defined prompts particularly on the block hybrid for BVPs and spectral block hybrid method for PDEs.
- **Facilitators:** Dr. Nosipho Zwane and Mr. Siyabonga Nozwane (South African Weather Service)

- **Theme:** The workshop introduces participants to the trends and opportunities for emerging research in climate modelling and its applications across sensitive sectors.
- **Facilitators:** Dr. Shina Daniel Oloniiju (Rhodes University).
- **Theme:** The workshop introduces participants to the theoretical understanding of block hybrid methods and their derivation to solve third-order and fourth-order boundary value problems.

Prof. Sandile Motsa

Dr. Shina Daniel Oloniiju

Dr. Nosipho Zwane

Mr. Siyabonga Nozwane

Workshop Facilitators

## 4 Workshop Schedule

| Time | Activity | Venue |
|---|---|---|
| | **Monday, 29 June 2026** | |
| 08:30–09:00 | Registration | G24 Lab |
| 09:00–09:10 | Welcome: Prof. Precious Sibanda | G24 Lab |
| 09:15–09:30 | Opening Address: Prof. Gueguim Kana | G24 Lab |
| 09:30–10:30 | Prof. Sandile Motsa: Generative AI in numerical methods. | G24 Lab |
| 10:30–11:00 | Tea Break | MSB F15 |
| 11:00–12:20 | Prof. Sandile Motsa: Block hybrid for BVPs - AI assisted derivation and solution. | G24 Lab |
| 12:20–13:00 | Groups: Exercise Session | G24 Lab |
| 13:00–14:00 | Lunch | MSB F15 |
| 14:00–15:50 | Prof. Sandile Motsa: Spectral block hybrid method for PDEs. | G24 Lab |
| 15:50–16:30 | Groups: Exercise Session | G24 Lab |
| | **Tuesday, 30 June 2026** | |
| 09:00–10:30 | Dr. Nosipho Zwane and Mr. Siyabonga Nozwane: Climate modelling in South Africa: Trends and opportunities for emerging research. | G24 Lab |
| 10:30–11:00 | Tea Break | MSB F15 |
| 11:00–13:00 | Group: Exercise Session | G24 Lab |
| 13:00–14:00 | Lunch | MSB F15 |
| 14:00–15:50 | Dr. Nosipho Zwane and Mr. Siyabonga Nozwane: Applications of climate modeling across climate-sensitive sectors. | G24 Lab |
| 15:50–16:30 | Groups: Exercise Session | G24 Lab |
| | **Wednesday, 01 July 2026** | |
| 09:00–10:30 | Dr. Shina D. Oloniiju: Hybrid block method for third-order BVPs. | G24 Lab |
| 10:30–11:00 | Tea Break | MSB F15 |
| 11:00–13:00 | Groups: Exercise Session | G24 Lab |
| 13:00–14:00 | Lunch | MSB F15 |
| 14:00–15:50 | Dr. Shina D. Oloniiju: Hybrid block method for fourth-order BVPs. | G24 Lab |
| 15:50–16:30 | Groups: Exercise Session | MSB F15 |

## 5 Conference Schedule

For accurate schedule updates, see `https://woccom.github.io/`.

**Thursday, 02 July 2026**

| | |
|---|---|
| Refreshments | Plenary Talks |
| Closing | Contributed Talks |

| Time | Activity | Venue |
|---|---|---|
| 08:15 – 09:00 | Registration | DSLT |
| 09:00 – 09:15 | Opening address: Prof. Precious Sibanda | DSLT |
| 09:15 – 10:00 | Plenary: Prof. Thando Ndarana<br>*Title: The dynamics of extratropical weather systems affecting Southern Africa*<br>Chair: Prof. Stanford Shateyi | DSLT |
| 10:00 – 10:30 | Tea break | MSB F15 |

| **Session 1:** | Prof. Melusi Khumalo and Mrs. Nancy Mukwevho | | | Prof. Sandile Motsa and Dr. Samah Ali | | |
|---|---|---|---|---|---|---|
| **Time** | **Name** | **Topic** | **Venue** | **Name** | **Topic** | **Venue** |
| 10:30 – 10:50 | Serai I. Mosala | *Entropy generation in steady MHD nanofluid Couette flow with partial slip and convective cooling: A spectral approach* | Council Chambers | Nefale Mpho | *Block hybrid methods for fourth-order boundary value problems* | DSLT |
| 10:50 – 11:10 | Ali Haroon | *Dynamic bilevel optimisation for carbon emission reduction: Integrating nonlinear policy functions and adaptive genetic algorithms* | Council Chambers | Lekau Letsoalo | *Optimal reinsurance and investment under Hawkes-type claim contagion and regime-dependent risk* | DSLT |

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| Time | Activity (Continued) | | | | | Venue |
|---|---|---|---|---|---|---|
| 11:10 – 11:30 | Abhijit Sharma | *Bioconvection and magnetohydrodynamic flow of trihybrid nanofluids containing motile microorganisms through porous media: A modified Buongiorno model approach* | Online | James Muila | *Multi-domain spectral collocation and semi-discrete schemes for solving Black-Scholes equation* | DSLT |
| 11:30 – 11:50 | Mapule Pheko | *Impact of Cattaneo–Christov fluxes on bio-convective flow of a second-grade hybrid nanofluid in a porous medium* | Council Chambers | Mokiri Nkwana | *Solving higher order Korte De-Vries (KdV) equations using multi-domain spectral collocation method* | DSLT |
| 11:50 – 12:10 | Kizito Ugochukwu Nwajeri | *Optimized spectral quasi linearization of unsteady MHD nanofluid squeezing flow with heat-mass transfer analysis* | Council Chambers | Robert Dozva | *A hybrid block method for linear and nonlinear systems of fractional differential equations* | DSLT |
| 12:10 – 12:30 | Nefale Phindulo | *On the numerical approach of MHD mixed convection flow and radiative heat transfer of Sisko nanofluid over nonlinear stretching sheet with viscous Ohmic dissipation* | Council Chambers | Ntandazo Mpisi | *An AI-Driven decision support system for intelligent selection and solution of differential equation models in scientific computing* | DSLT |
| 12:30 – 12:50 | Zamachube Chamane | *Mathematical modelling of malaria transmission dynamics incorporating immunity dynamics and control interventions* | Council Chambers | Mulweli Mudau | *An overlapping multi-domain bivariate simple iteration method for nonlinear partial differential equations* | DSLT |
| 12:50 – 13:00 | Photo session | | | | | Outside Council Chambers |
| 13:00 – 14:00 | Lunch | | | | | MSB |
| **Session 2:** Dr. Sivuyile Mgobhozi and Dr. Nosipho Zwane | | | | Dr. Anastacia Dlamini and Dr. Kizito Muzhinji | | |
| **Time** | **Name** | **Topic** | **Venue** | **Name** | **Topic** | **Venue** |

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| Time | Activity (Continued) | | | | | Venue |
|---|---|---|---|---|---|---|
| 14:00 – 14:20 | Anubhab Bhandari | *MHD flow of ternary nanofluids over a rotating porous disk with interfacial morphology effects* | Online | Thabo Tsebe | *Mathematical modelling and high-order numerical analysis of emerging HIV prevention strategies in Limpopo Province, South Africa* | DSLT |
| 14:20 – 14:40 | Samriddha Deb | *Entropy analysis of cross trihybrid nanofluid flow with nanolayer thickness using the overlapping grid method* | Online | Vhutshilo Molaudzi | *Comparative performance analysis of a block-hybrid method and fourth-order fourth-order Runge-Kutta method for solving first-order ordinary differential equations* | DSLT |
| 14:40 – 15:00 | Boineelo Mpheng | *Mathematical modelling and computational analysis of two-phase channel flows* | Council Chambers | Daniel Mahlangu | *Nonlinear interaction of large-amplitude plasma solitary waves in the small amplitude-difference regime* | DSLT |
| 15:00 – 15:20 | Sinenhlanhla Hlope | *Spectral collocation methods for pollution diffusion with advection* | Council Chambers | Sibonelo Nzama | *A comparative study of trivariate spectral collocation and overlapping grids multidomain spectral-linear partition approaches* | DSLT |
| 15:20 – 15:40 | Zwelithini Mkhonta | *Optimized overlapping and non-overlapping block hybrid methods for nonlinear second-order boundary value problems* | Council Chambers | Annah Nivia Malatjie | *American put pricing option under stochastic interest rates and finite time horizon* | DSLT |
| 15:40 – 16:00 | David Moseki | *Sensitivity analysis for nanofluid flow driven by thermal and mass diffusion through a rotating convergent/divergent channel* | Council Chambers | Shamone Ruiters | *Nonstandard finite difference schemes for differential equation in combustion theory* | DSLT |
| 16:00 – 16:20 | Siphelele Mbanguta | *A compartmental model for the epidemiology of depression: formulation, dynamics, and implications* | Council Chambers | Lekaba Thabo Maomela | *Enhanced solution for the advection–diffusion–reaction equation using the Physics-Informed Neural Network technique* | DSLT |

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| **Time** | **Activity (Continued)** | **Venue** |
|---|---|---|
| 18:00 – 22:00 | Dinner | Golden Horse Casino |

**Friday, 03 July 2026**

| Time | Activity | Venue |
|---|---|---|
| 08:30 – 09:15 | Plenary: Dr. Olumuyiwa Otegbeye<br>*Title: Neural network solution of differential equations*<br>Chair: Prof. Melusi Khumalo | DSLT |
| 09:15 – 10:00 | Plenary: Prof. Montaz Ali<br>*Title: A linear predictor of a non-linear dynamical systems with applications*<br>Chair: Prof. Lazarus Rundora | DSLT |
| 10:00 – 10:30 | Tea break | MSB F15 |

**Session 3:** Dr. Letlhogonolo Moleleki — Prof. Precious Sibanda

| Time | Name | Topic | Venue | Name | Topic | Venue |
|---|---|---|---|---|---|---|
| 10:30 – 10:50 | Lesiba Charles Galane | *Stochastic differential reinsurance game formulation and investment problem under power, logarithmic and exponential utility functions* | Council Chambers | Phangisile Khumalo | *Thermal analysis of functionally graded porous fins with spatially varying conductivity and porosity using a spectral collocation approach* | DSLT |
| 10:50 – 11:10 | Amna Ibrahim | *Stability analysis and optimal control of a Hepatitis C Virus (HCV) model with immune response* | Council Chambers | Rendani Netshikweta | *Modeling foot-and-mouth disease dynamics among livestock and wild ruminants: Integrating community viral load and environmental transmission pathways* | DSLT |
| 11:10 – 11:30 | Zakaria Ali | *Application of the block hybrid method to nonlinear heat conduction in a human head model* | Council Chambers | Juliet Molepo | *Two-phased unsteady second grade tetra hybrid nanofluid flow through stretching Riga surface with microgravity: Overlapping grid based spectral collocation method* | DSLT |
| 11:30 – 11:50 | Gabriel Simaneka Mbokoma | *A meshless RBF-FD solver for compressible Euler's equation with shock-sensor based artificial viscosity* | Council Chambers | Tshepho Chukudu | *Mathematical modelling of tumor growth dynamics and metastatic spread: McKendrick-von Foerster approach* | DSLT |

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| Time | Activity (Continued) | | | | | Venue |
|---|---|---|---|---|---|---|
| 11:50 – 12:10 | Siyamthanda Gift Mnisi | *Southern African HIV/AIDS dynamics analysis integrating statistical and mathematical modelling* | Council Chambers | Phetulo Sedibe | *Lie symmetry analysis of the Lane-Emden equation* | DSLT |
| 12:10 – 12:30 | Interaction and Networking among Participants | | | | | DSLT |
| 12:30 – 12:50 | Presentation of Prizes and Awards | | | | | DSLT |
| 12:50 – 13:00 | Closing | | | | | DSLT |
| 13:00 – 14:00 | Lunch | | | | | MSB |

**Student Evaluators**

- Prof. Zakaria Ali (University of South Africa)
- Dr. Musawenkosi Mkhatshwa (University of Limpopo)
- Dr. Thama Duba (University of Witwatersrand)
- Dr. Lesiba Charles Galane (University of Limpopo)
- Dr. Nosipho Zwane (South African Weather Service)
- Dr. Letlhogonolo Moleleki (North-West University)
- Dr. Hloniphile Sithole-Mthethwa (University of KwaZulu-Natal)
- Dr. Vusi Magagula (University of Eswatini)
- Dr. Ndivhuwo Ndou (University of Venda)

# 6 Plenary Abstracts

### The dynamics of extratropical weather systems affecting Southern Africa

**Thando Ndarana** (*University of Pretoria*)

Idealized modelling studies have shown that baroclinic life cycles (BLCs), which begin with the vertical propagation of wave activity in the extratropics and its subsequent absorption or reflection in the subtropics as baroclinic waves dissipate, are critical for understanding the dynamics of extratropical weather systems. During the dissipation stage of BLCs, Rossby wave breaking (RWB) occurs and manifests as the rapid and irreversible overturning of potential vorticity contours. This RWB leads to the formation of important weather systems, such as cut-off low-pressure systems, which may occur over Namibia, Botswana, and South Africa. These systems are often associated with the ridging of the South Atlantic anticyclone at the lower levels, which is responsible for transporting moisture from the southwest Indian Ocean onto the southern african mainland. This moisture may extend as far into the subcontinent as Botswana.

In this presentation, we discuss these dynamical processes from the perspective of energy transfer, beginning with baroclinic conversion over the South Atlantic Ocean and the associated downstream development that eventually leads to the formation of cut-off lows and other weather systems. Developing studies at the University of Pretoria that address these processes directly from a finite-amplitude wave activity perspective will also be discussed in order to make a case for the BLC framework as a useful approach for understanding the weather and climate of southern Africa.

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### A linear predictor of a non-linear dynamical systems with applications

**Montaz Ali** (*University of the Witwatersrand*)

This talk presents a new dynamic mode decomposition (DMD)-an equation-free technique for predicting the system dynamics. The technique does not require to know the underlying governing equations where the finite dimensional approximation of the Koopman operator plays a fundamental role. Our predictor is more general than its original counterpart. Unlike the original DMD which uses the pseudo-inverse technique for constructing the linear operator, we have used constrained optimization technique.

The new method proves its superiority over the original over longer time horizon. A control parameter is then introduced in the DMD as an interventional strategy for paediatric growth monitoring. The linear operator and the parameters of the control are obtained from data generated from a neural ode while the control values are obtained under the model predictive strategy. Both model have been validated using test cases.

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### Neural network solution of differential equations

**Olumuyiwa Otegbeye** (*University of the Witwatersrand*)

Traditional numerical methods provide rigorous solutions to differential equations but face steep computational costs when dealing with complex geometries or high-dimensional spaces. Neural networks particularly Physics-Informed Neural Networks (PINNs) offer a promising mesh-free alternative, however, they frequently struggle with stiff equations, high-frequency oscillations, and the strict enforcement of boundary conditions.

This talk presents a novel framework that addresses these deep learning limitations without discarding the rigor of classical mathematics. We introduce some novelty into the neural network architecture that dynamically improves the performance of the neural network algorithm. This approach effectively mitigates the spectral bias of standard networks, allowing the model to learn smooth macro-scale dynamics and sharp, complex physical features simultaneously.

Through comparative examples of linear and non-linear differential equations, we will demonstrate how this optimized method achieves faster convergence and higher accuracy than baseline neural solvers. Ultimately, this presentation highlights a practical step forward: blending the mathematical reliability of traditional numerical constraints with the flexibility and speed of modern machine learning.

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## 7 Contributed Abstracts

### A compartmental model for the epidemiology of depression: Formulation, dynamics, and implications

**Siphelele Mbanguta** (*Sefako Makgatho Health Sciences University*)

This paper introduces an original compartmental model for analyzing the spread of Major Depressive Disorder (MDD) in populations using deterministic methods. The model divides the population into seven compartments: Susceptible (S), Untreated and undiagnosed (U), Primary/EarlySigns (P), Chronic (D), Early Non-Pharmaceutical Intervention (M), Chronic Medication (C), and Recovered (R). The model incorporates social transmission of depressive symptoms, clinical progression, treatment pathways, recovery, relapse, and mortality.
All parameters are empirically grounded using South African longitudinal data (NIDS, SADHS) and social influence weights derived from the Add Health study. The basic reproduction number is calculated as $R_0 \approx 1.50$, indicating that depression is endemic under current conditions and that a 33% reduction in transmission could achieve elimination. Global stability analysis using Lyapunov functions demonstrates that the disease-free equilibrium is globally asymptotically stable when $R_0 < 1$, and the endemic equilibrium is globally asymptotically stable when $R_0 > 1$. Sensitivity analysis reveals that early intervention parameters—particularly the therapy initiation rate ($\theta_p$) and diagnosis rate ($\alpha$) exert the strongest negative influence on $R_0$. The framework provides a simulation platform for evaluating depression control measures and analyzing community-level disease spread, with direct applications to public health policy.
**Keywords:** Analysis, Lyapnov stability.

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## Entropy generation in steady MHD nanofluid Couette flow with partial slip and convective cooling: A spectral approach

**Serai Israel Mosala** (*Nelson Mandela University*)

This study investigates the steady-state thermal and magnetohydrodynamic (MHD) behaviour of electrically conducting nanofluids in a Couette flow configuration between two infinite parallel plates. The upper plate moves at a constant velocity under partial slip conditions, while convective heat exchange is imposed at the upper boundary and a fixed temperature is maintained at the lower plate. Cu-Water and $Al_2O_3$-Water nanofluids are considered, modelled using a single phase continuum framework in which effective thermophysical properties including density, viscosity, thermal conductivity, heat capacity, and electrical conductivity are expressed as functions of nanoparticle volume fraction. The governing momentum, energy, and entropy generation equations are non-dimensionalised and solved using the Spectral Quasi-Linearisation Method (SQLM), which offers rapid convergence for nonlinear boundary value problems. Parametric analysis is conducted across key dimensionless groups: the Hartmann number (Ha), Eckert number (Ec), nanoparticle volume fraction ($\phi$), slip parameter ($\lambda$), pressure gradient parameter (G), Reynolds number (Re), and Biot number (Bi). Results show that increasing the Hartmann number suppresses fluid velocity through the opposing Lorentz force while amplifying temperature gradients via Joule dissipation. Higher nanoparticle volume fractions enhance thermal conductivity and improve heat transfer rates as reflected in increasing Nusselt numbers, though with increased flow resistance. Entropy generation and Bejan number profiles are examined to quantify the relative contributions of heat transfer irreversibility, viscous dissipation, and magnetic field effects to overall thermodynamic inefficiency. The findings offer fundamental insights applicable to thermal management systems, electromagnetic cooling devices, and micro-electromechanical applications.

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## Dynamic bilevel optimisation for carbon emission reduction: Integrating nonlinear policy functions and adaptive genetic algorithms

**Ali Haroon** (*University of KwaZulu-Natal*)

This work presents a dynamic bilevel optimisation framework for designing and evaluating carbon emission reduction policies. It models the strategic interaction between a policymaker (the leader), who seeks to minimize emissions, and profit-maximizing energy producers (the followers). We introduce nonlinear tax and subsidy functions that evolve over time, enabling a more realistic simulation of escalating policy interventions. Energy production technologies are classified into three categories: high-carbon, medium-carbon, and low-carbon, based on their carbon intensity and unit production costs. Our theoretical analysis establishes the existence of a solution and, most importantly, formally proves that these policy instruments can guarantee a finite-horizon phase-out of high-carbon sources. To solve the model, we propose an adaptive genetic algorithm with time-decaying crossover and mutation parameters. We validate the framework in a case study calibrated with data from South Africa's energy sector. The results show that the optimal dynamic policy substantially outperforms static benchmarks, achieving deep decarbonization by engineering a predictable coal phase-out while satisfying system constraints. Our framework offers a prescriptive and robust decision support tool for policymakers to design and calibrate adaptive, data-driven climate policies.

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## Thermal analysis of functionally graded porous fins with spatially varying conductivity and porosity using a spectral collocation approach.

**Phangisile Khumalo** (*University of KwaZulu-Natal*)

Functionally graded materials (FGMs) have attracted considerable attention due to their enhanced thermal performance in advanced heat transfer applications. However, the coupled effects of spatially

varying thermal conductivity and surface porosity on fin performance, particularly under transient conditions, remain insufficiently explored. This study investigates the thermal behaviour of longitudinal FGM fins by analysing the combined influence of material grading, exponential porosity decay, radiation, and convective heat transfer under steady-state and transient conditions. A nonlinear heat transfer equation is formulated, nondimensionalised, and solved numerically using the spectral collocation method (SCM) for steady-state analysis and the bivariate spectral collocation method (BSCM) for transient analysis, considering linear, quadratic, exponential, and power-law conductivity profiles. The results reveal that increased surface porosity reduces heat dissipation by decreasing the effective heat transfer area and increasing internal thermal resistance, leading to higher temperature distributions. Linear, quadratic, and exponential conductivity profiles increase temperature, whereas the power-law profile enhances thermal performance by redistributing conductivity away from the base. Validation with analytical solutions and finite element method (FEM) simulations shows excellent agreement. Parametric analysis indicates that radiation and higher Biot numbers improve heat dissipation but reduce fin efficiency, while internal heat generation intensifies thermal gradients. These findings provide a practical framework for designing high-efficiency, lightweight thermal systems.

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## Mathematical modelling and computational analysis of two-phase channel flows.

**Boineelo Mpheng** (*University of Limpopo*)

This study presents a mathematical and computational investigation of two-phase channel flows, which include the fluid–dust particle systems in various channel geometries. This work is motivated by the industrial application of multiphase flows in biomedical, petroleum and energy systems. This research addresses existing gaps in modelling multi-fluid interactions, non-Newtonian behaviour, and hybrid nanoparticle effects. Mathematical models that include continuity, momentum, energy, and concentration equations for each phase are formulated, incorporating interfacial boundary conditions. The finite difference and improved multi-domain spectral collocation methods are used to solve the dimensionless equations. These numerical methods are capable of handling strongly coupled nonlinear systems with internal interface conditions. Response surface methodology and sensitivity analysis are employed to optimize key thermophysical parameters governing heat and mass transfer. The influence of physical parameters on flow fields, shear stress, entropy generation and transport quantities are systematically examined. The obtained numerical results are validated against MATLAB pvp4c solver and existing solutions for limiting cases.

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## Mathematical modelling and high-order numerical analysis of emerging HIV prevention strategies in Limpopo Province, South Africa

**Thabo Tsebe** (*University of Limpopo*)

This study develops a mathematical model to analyse HIV transmission dynamics and evaluate the combined impact of prevention strategies in Limpopo Province, South Africa. The model integrates key interventions, including oral and injectable pre-exposure prophylaxis (PrEP), treatment as prevention (TasP), HIV testing, and behavioural changes. It incorporates realistic features such as adherence variability, pharmacological tail effects, and risk compensation. Qualitative analysis is performed to determine model properties and stability, with the basic reproduction number derived using the next-generation matrix method. An optimal control framework is used to identify cost-effective intervention strategies. A high-order Block Hybrid numerical method is developed and compared with the classical Runge–Kutta method for improved accuracy and efficiency. Numerical simulations and sensitivity analysis performed using MATLAB provide valuable insights into the most effective strategies for reducing HIV transmission in resource-limited settings.

**Keywords:** HIV transmission, mathematical modelling, PrEP, injectable PrEP, TasP optimal control, numerical methods, Block Hybrid method, Runge–Kutta method, adherence, Limpopo Province.

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## Multi-domain spectral collocation and semi-discrete schemes for solving Black-Scholes equation

**James Muila** (*University of Limpopo*)

Black-Scholes equation is an important mathematical model used in finance for pricing options such as European call and put options. Even though exact solutions exist under simple assumptions, many practical cases, especially nonlinear forms, are difficult to solve analytically. Because of this, numerical methods such as finite difference and spectral collocation methods are commonly used. However, they may face challenges because of non-smooth payoffs, unbounded domains, large computational domains, and backward time direction in the Black-Scholes equation. Therefore, this study proposes the use of multi-domain spectral collocation and semi-discrete schemes for solving Black-Scholes equations. The Black-Scholes models is first transformed into diffusion-type equations using suitable variable transformations to simplify the problem and improve numerical stability. The unbounded asset price domain is reduced to a finite interval, while the spatial domain is divided into overlapping subintervals to improve accuracy near important regions such as the strike price. After spatial discretization, time integration is carried out using various techniques, including non-overlapping multi-domain spectral collocation, implicit time stepping $\theta$-methods, and Backward Differentiation Formula. All numerical algorithms are implemented in MATLAB. The efficiency, accuracy, stability, and convergence of the proposed methods are compared with existing classical techniques.

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## Two-phased unsteady second grade tetra hybrid nanofluid flow through stretching Riga surface with microgravity: Overlapping grid based spectral collocation method

**Juliet Molepo** (*University of Limpopo*)

This study examines the effect of dusty particles on an unsteady second-grade fluid with suspension of tetra hybrid nanofluid through stretching Riga surface along microgravity conditions. The governing partial differential equations are transformed into non-dimensional form using suitable transformations. The obtained non-dimensional equations are solved numerically using the overlapping-grid spectral collocation method implemented in MATLAB. The results of varying parameters are presented in graphics and tabular form, with validation through comparison with existing literature. The current flow model plays a crucial role in biomedical applications, cooling systems, and solar power generation.

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## Solving higher order Korte De-Vries (KdV) equations using multi-domain spectral collocation method

**Mokiri Mathews Nkwana** (*University of Limpopo*), Musawenkosi Mkhatshwa, Kgomotshwana Thosago

Solving KdV equations has been Long-standing challenge particularly for large computational domains. This study introduces the multi-domain spectral collocation method for solving higher order Korteweg De Fries (KdV) equations defined on large time frames. The KdV equations are firstly altered to a linearized form of iterative scheme using quasi-linearization method. The time domain is decomposed into non-overlapping sub-intervals, while on the contrary space domain is disintegrated into overlapping sub-intervals. The solutions at each time sub-interval are computed independently, where continuity condition is applied to produce initial conditions for the subsequent subintervals, whereas the solutions

in the space interval are to be determined simultaneously across overlapping sub-intervals. The solutions are correlated with the exact solutions to manifest accuracy. Furthermore, the efficacy, stability and accuracy of the method are demonstrated by presenting computational error analysis, condition numbers and the computational time for the solution of KdV equations.The method has proven to be a reliable numerical tool, capable of accurately capturing the intricate structures typical of KdV type models, such as solitons and complex wave interactions. The method also demonstrated high accuracy, with numerical solutions closely matching available exact solutions, yielding small absolute errors.
**Keywords:** KdV equation, Quasi-linearization method, Multidomain spectral collocation method, bi-variate Lagrange interpolation

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## Spectral collocation methods for pollution diffusion with advection

**Sinenhlanhla Hlope** (*University of KwaZulu-Natal*)

This study investigates pollutant transport in two distinct water systems, a one-dimensional (1D) stream flow and a two-dimensional (2D) flow inside a cylindrical pipe, by solving two different advection–diffusion equations (ADEs) under various physical conditions. The pollutant stream flow model is analysed in two cases. Case 1 excludes source and reaction terms to establish a simplified baseline, allowing the effects of flow speed (advection) and diffusion coefficient to be computationally examined. Case 2 incorporates both a pollutant source and a chemical reaction term to assess their combined influence on the dynamics observed in case 1. The cylindrical pipe model addresses a purely advection–diffusion equation (ADE) system, enabling the study of pollutant concentration evolution under different radial and axial flow speed at constant diffusion coefficient. To ensure the reliability and accuracy of the results obtained from these models, two numerical methods are implemented, the Bivariate Spectral Collocation Method (BSCM) for the stream flow system, and the Trivariate Spectral Collocation Method (TSCM) for the cylindrical pipe system. Their application enables efficient computation of the solution while maintaining numerical stability. The methods also support residual-based validation, which confirms that the computed results remain consistent, convergent, and physically meaningful across a wide range of parameter settings. The results highlight how flow speed, diffusion strength, pollutant source and chemical reactions affect the concentration of the pollutant as it propagates through the water flow system. Overall, this study provides a robust computational framework for simulating pollutant transport and reinforces the reliability of the findings under diverse environmental conditions. The insights obtained can be applied to inform mitigation strategies and support environmental risk assessment.

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## Application of the block hybrid method to nonlinear heat conduction in a human head model

**Zakaria Ali** (*University of South Africa*)

This study investigates the complex spatio-temporal distribution of temperature within the human head under conditions of elevated metabolic heat production and external thermal stress. By employing a modified Pennes' Bio heat equation, we analyze the conditions under which a bio-thermal system may undergo a thermal explosion which is a state of runaway temperature increase that leads to irreversible cellular damage. We establish rigorous stability criteria using the Frank-Kamenetskii theory, deriving a critical parameter $\delta_c$ that defines the boundary between stable homeostasis and thermal instability. Our findings provide a mathematical foundation for understanding extreme hyperthermal pathologies and suggest critical thresholds for neuro-thermal safety.

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### Multi-domain spectral discretization approach for solving time-fractional KdV equations

**Musawenkosi Mkhatshwa** (*University of Limpopo*)

This study introduces an innovative numerical approach to solve nonlinear time-fractional Korteweg–de Vries (KdV) equations. These KdV equations, fundamental in describing nonlinear wave phenomena, present complexities due to the involvement of fractional derivatives. In the numerical approach, overlapping Chebyshev spectral collocation (SC) is utilized for spatial discretization and non-overlapping multi-domain SC is applied for time discretization, where the Caputo-type derivative is approximated in each time sub-interval and the resulting intricate integral is approximated using quadrature rule. Furthermore, the quasi-linearization technique is utilized to handle the nonlinear terms and the derivatives of the temporal Lagrange basis functions are evaluated using the barycentric differentiation formula. The non-overlapping temporal decomposition and SC in time provide superior accuracy, enhanced numerical stability, and systematic treatment of the nonlocal time-fractional derivative through sub-interval-wise memory contributions. In addition to those benefits, the overlapping spatial decomposition and SC in space enhance the local resolution, minimize the size of Chebyshev differentiation matrices, and improve the treatment of complicated wave structures. The obtained results underscore the efficacy of this novel numerical method in handling complex nonlinear equations involving fractional derivatives, suggesting its potential for broader applicability in similar mathematical problems.

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### Modeling foot-and-mouth disease dynamics among livestock and wild ruminants: Integrating community viral load and environmental transmission pathways

Mukhethwa Chantel Kaletsane, Azwindini Delinah Maphiri, **Rendani Netshikweta** (*University of Limpopo*)

Foot-and-mouth disease (FMD) is a highly transmissible viral infection of livestock that threatens food security and causes substantial economic losses in endemic regions. Despite its economic impact, the role of environmental viral load and wildlife reservoirs in sustaining FMD transmission remains poorly quantified. The aim of this study is to assess the extent to which community viral load sustains FMD persistence and to identify key transmission drivers in a coupled livestock–wildlife–environment system. A Susceptible–Exposed–Infected (SEI) model with a free-living virus compartment was analyzed via the basic reproduction number (R0) and solved numerically using a Nonstandard Finite Difference Method. Sensitivity analysis identified wild host population size, transmission rates, host recruitment, environmental viral decay, and viral load thresholds as major determinants of R0. Results indicate that higher transmission rates accelerate susceptible depletion and increase exposed and infected classes, with wildlife dominating environmental viral contributions. Community viral load is central to sustaining outbreaks and informs targeted control strategies.
**Keywords:** Foot-and-mouth disease (FMD), mathematical modeling, wild animals, livestock, community viral load, nonstandard finite difference method.

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### Stability analysis and optimal control of a Hepatitis C Virus (HCV) model with immune response

**Amna Ibrahim** (*University of KwaZulu-Natal*)

This study presents a comprehensive mathematical framework to analyze the immunopathogenesis and optimal treatment strategies for Hepatitis C Virus (HCV) infection. First, we develop a system of nonlinear differential equations that captures the complex interactions among hepatocytes, HCV,

immune cells, and specific cytokines. We establish the well-posedness of the model, calculate the basic reproduction number ($R_0$), and derive conditions for the stability of disease-free and endemic equilibria. Sensitivity analysis and numerical simulations under varying immune scenarios reveal that T-helper cells, Interleukin-2 ($IL-2$), are crucial for cytotoxic T lymphocyte ($CTL$) activation and accurate disease dynamics. Building upon this biological foundation, treatments are incorporated into the model to determine optimal control strategies. Using Pontryagin's Maximum Principle, we aim to minimize the viral load and drug toxicity. A comparative analysis of interferon monotherapy, ribavirin monotherapy, and combination therapy demonstrates that combination treatment achieves the most effective viral suppression, the shortest treatment duration, and the lowest overall side effects.

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## Impact of Cattaneo–Christov fluxes on bio-convective flow of a second-grade hybrid nanofluid in a porous medium

**Mapule Pheko** (*North-West University*)

The study investigates the flow of a second-grade hybrid nanofluid through a Darcy–Forchheimer porous medium under Cattaneo–Christov heat and mass flux models. The hybrid nanofluid, composed of alumina and copper nanoparticles in water, enhances thermal and mass transport, while the second-grade model captures viscoelastic effects, and the Darcy–Forchheimer medium accounts for both linear and nonlinear drag. Using similarity transformations and the spectral quasilinearisation method, the nonlinear governing equations are solved numerically and validated against benchmark results. The results show that hybrid nanoparticles significantly boost heat and mass transfer, while Cattaneo–Christov fluxes delay thermal and concentration responses, reducing the near-wall temperature and concentration. The distributions of velocity, temperature, concentration, and microorganism density are markedly affected by porosity, the Forchheimer number, the bio-convection Peclet number, and relaxation times. The results illustrate that hybrid nanoparticles significantly increase heat and mass transfer, whereas thermal and concentration relaxation factors delay energy and species diffusion, thickening the associated boundary layers. Viscoelasticity, porous medium resistance, Forchheimer drag, and bio-convection all have an influence on flow velocity and transfer rates, highlighting the subtle link between these mechanisms. These breakthroughs may be beneficial in establishing and enhancing bioreactors, microbial fuel cells, geothermal systems, and other applications that need hybrid nanofluids and non-Fourier/non-Fickian transport.

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## Comparative performance analysis of a block hybrid method and fourth-order fourth-order Runge-Kutta method for solving first-order ordinary differential equations

**Vhutshilo Molaudzi** (*University of Venda*)

This study compares the Block Hybrid Method (BHM) and the classical fourth-order Runge-Kutta (RK4) method for solving first-order ordinary differential equations (ODEs), including linear, nonlinear, and epidemiological models. Effective numerical methods are required to obtain accurate approximations because many of these problems lack analytical solutions. The proposed BHM computes multiple solution points simultaneously within a single integration block, thereby improving computational efficiency. To handle nonlinear systems, the quasilinearization technique is used to transform the nonlinear ODEs into a sequence of linear problems that can be solved iteratively. MATLAB software was used for each and every numerical simulation. All numerical simulations were performed using MATLAB. Numerical results show that the BHM produces significantly smaller absolute errors than the RK4 method while maintaining excellent agreement with the exact solutions. In addition, the BHM demonstrates improved computational efficiency through reduced CPU times and favorable stability behaviour.

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## Optimal reinsurance and investment under Hawkes-type claim contagion and regime-dependent risk

**Lekau Letsoalo** (*University of Limpopo*)

This study develops an optimal reinsurance and investment model for an insurer facing claim contagion and macroeconomic uncertainty. Insurance claims are modelled using a Hawkes process to capture self-exciting and clustered losses, while economic conditions are represented by a finite-state Markov regime-switching process. The insurer aims to maximise expected terminal utility through optimal reinsurance and dynamic investment in risky and risk-free assets under regime-dependent market and insurance parameters. Using stochastic control theory, the problem is formulated as a Hamilton-Jacobi-Bellman system. Due to the model's complexity, numerical methods are employed to analyse optimal strategies. This framework extends classical insurance models by incorporating both claim clustering and changing economic conditions, providing a more realistic approach to risk management and solvency planning.

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## Stochastic differential reinsurance game formulation and investment problem under power, logarithmic and exponential utility functions

**Lesiba Charles Galane** (*University of Limpopo*)

This talk presents a stochastic differential game between two competing insurance companies that simultaneously engage in reinsurance and investment in risky financial assets. Each insurer maximizes the expected utility of terminal wealth under exponential, logarithmic, and power utility functions, reflecting heterogeneous risk preferences. The surplus processes are modelled by correlated Brownian motions, thereby capturing interdependence in both underwriting risk and financial markets. The Hamilton-Jacobi-Bellman (HJB) and Fleming-Bellman-Isaacs (FBI) equations are used to characterize the strategic interaction, and Nash equilibrium strategies are derived, ensuring that no insurer can unilaterally improve its position. The results highlight the critical influence of correlation between surplus processes: stronger correlation amplifies competition and limits diversification, while weaker correlation allows more aggressive investment strategies. Simulation results further show that insurers with greater initial reserves retain more risk and invest heavily in risky assets, while weaker competitors rely on reinsurance for protection. The study extends existing literature by jointly considering risky investment, surplus correlation, and heterogeneous risk preferences, offering theoretical contributions and practical insights for insurers operating in competitive and uncertain markets.

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## A hybrid block method for linear and nonlinear systems of fractional differential equations

**Robert Dozva** (*University of KwaZulu-Natal*), Sicelo P. Goqo, Shina D. Oloniiju

The non-local behavior of fractional differential equations (FDEs) creates major difficulties when trying to obtain closed-form solutions and, by extension, numerical solutions for systems that include coupled equations used to model complex phenomena. In the literature, a considerable number of numerical methods have been developed for fractional differential equations (FDEs) and for systems of FDEs. However, most of these methods exhibit deficiencies in achieving high accuracy, high efficiency, and exact reproduction of the memory effects at the same time. This paper presents a high-order one-step hybrid block method (FHBM) for solving systems of FIVPs with Caputo fractional derivative of order $\alpha \in (0, 1]$. The development of the method utilizes Lagrange interpolation to construct a block system that achieves multiple point approximations through equispaced intra-step collocation points. The method extends its ability to handle complex systems through vectorization of the scalar scheme which achieves high

accuracy without requiring large computational power for different system dimensions. The method achieves zero stability according to theoretical analysis while producing a local truncation error of order $\mathcal{O}(h^{M+1+\alpha})$. The proposed method outperforms standard finite difference and extrapolation schemes from the literature according to numerical tests conducted on linear and non-linear chaotic fractional systems. The method achieves quick convergence rates through its results obtained for different values of $M$, the intra-step points while it maintains stability for stiff problems that contain interconnected equations.
**Keywords:** Fractional differential equations, systems of FDEs, hybrid block methods, Chaotic systems, Lagrange interpolation.

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## American put pricing option under stochastic interest rates and finite time horizon

**Annah Nivia Malatjie** (*University of Limpopo*)

In this research, we study the pricing of American put options on a non-dividend-paying stock under a stochastic interest rate with a finite time horizon. We prove that the option value is a continuously differentiable function of time, interest rate and stock price. By means of Ito calculus, we rigorously derive the option values, the early exercise premium formula, and the associated hedging portfolio.
We prove the existence of an optimal exercise boundary splitting the state space into continuation and stopping region. The boundary has a parametrisation as a jointly continuous function of time and stock price and it is the unique solution to an integral equation which we compute numerically.
**Keywords:** American put option, stochastic interest rate, Vasicek model, free boundary problems

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## Optimized spectral quasi-linearization of unsteady MHD nanofluid squeezing flow with heat-mass transfer analysis

**Kizito Ugochukwu Nwajeri** (*University of KwaZulu-Natal*)

This study presents an optimized analysis of unsteady magnetohydrodynamic (MHD) squeezing flow of a nanofluid confined between two parallel plates, incorporating coupled heat and mass transfer effects. Using an appropriate similarity transformation, the governing nonlinear partial differential equations are reduced to a system of coupled nonlinear ordinary differential equations, which are solved numerically via a spectral quasi-linearization method exhibiting rapid convergence. The proposed numerical framework is validated against existing literature and classical shooting techniques, demonstrating excellent agreement and confirming the accuracy and robustness of the approach. The effects of key dimensionless parameters on the velocity, temperature, and concentration fields are systematically examined. In addition, relevant engineering quantities, including the skin friction coefficient, Nusselt number, and Sherwood number, are evaluated to characterize momentum, heat, and mass transfer rates. To further enhance predictive capability and enable parametric optimization, Response Surface Methodology (RSM) based on a Central Composite Design (CCD) is employed to construct surrogate models and to investigate interaction effects among the governing parameters. Sensitivity analysis and analysis of variance (ANOVA) results reveal that the heat transfer coefficient is most strongly influenced by the thermal radiation parameter, while the mass transfer coefficient is predominantly affected by the Lewis number. High predictive accuracy is achieved, with coefficients of determination of $R^2 = 0.9877$ (adjusted $R^2 = 0.9767$) for the heat transfer coefficient and $R^2 = 0.9860$ (adjusted $R^2 = 0.9733$) for the mass transfer coefficient. The integrated numerical–statistical framework provides deeper insight into the underlying transport mechanisms and offers a reliable tool for the analysis and optimization of MHD nanofluid systems in confined geometries. These findings are relevant to advanced thermal management systems, microfluidic devices, and energy-related transport processes.

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## MHD flow of ternary nanofluids over a rotating porous disk with interfacial morphology effects

**Anubhab Bhandari** (*North-Eastern Hill University, Shillong*)

This study investigates the heat and mass transfer characteristics of a ternary hybrid nanofluid flowing between two rotating porous disks under the influence of a magnetic field. The analysis incorporates several important physical effects, including Soret–Dufour diffusion, thermal radiation, viscous dissipation, inclined magnetic field, Darcy–Forchheimer resistance, and chemical reactions. The governing partial differential equations are transformed into a system of ordinary differential equations using appropriate similarity transformations and are numerically solved through the Spectral Quasi-Linearization Method (SQLM) combined with Chebyshev collocation. In addition, an Artificial Neural Network (ANN) model is developed to predict key flow quantities such as the skin friction coefficient, Nusselt number, and Sherwood number. This approach enables fast and accurate predictions without repeatedly solving the complete mathematical system. The results reveal that an increase in magnetic field inclination enhances radial and tangential velocities and thickens both thermal and concentration boundary layers. Higher Reynolds numbers strengthen convective transport and improve heat transfer, whereas chemical reactions reduce nanoparticle concentration. Moreover, thermal radiation and Soret–Dufour effects significantly affect temperature and concentration distributions. Overall, the study demonstrates that the combined application of SQLM and ANN provides accurate numerical solutions along with efficient predictive capability. The findings highlight the important roles of magnetic orientation, porous resistance, and nanoparticle composition in governing flow behavior and heat–mass transfer. These results offer useful guidance for the design of advanced thermal management systems, rotating disk devices, and energy applications involving ternary hybrid nanofluids.
**Keywords:** Ternary hybrid nanofluid, Magnetohydrodynamics (MHD), Rotating porous disks, Darcy–Forchheimer resistance, Thermal radiation, Heat and mass transfer, SQLM, Artificial neural network.

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## Mathematical modelling of malaria transmission dynamics incorporating immunity dynamics and control interventions

**Zamachube Chamane** (*University of KwaZulu-Natal*)

Malaria control efforts in Sub-Saharan Africa continue to face challenges arising from nonlinear transmission dynamics associated with reinfection, waning immunity and heterogeneous disease progression. In this study, a nonlinear malaria transmission model incorporating immunity feedback and control interventions is developed to investigate how immunity dynamics influence transmission thresholds and long-term control outcomes. The model divides the human population into epidemiological classes representing primary infection, temporary partial immunity, reinfection with enhanced post-infection immunity and eventual waning of protection back to full susceptibility, while the mosquito population is described using a susceptible–exposed–infectious framework. Analytical investigations include the derivation of the basic reproduction number (R-naught) equilibrium analysis and stability analysis of the disease-free and endemic steady states. Particular attention is given to immunity-driven threshold behavior and the potential emergence of nonlinear phenomena such as backward bifurcation, where disease persistence may occur even when transmission thresholds are below unity. Numerical simulations are used to examine the impact of intervention strategies under varying immunity and reinfection scenarios. The findings suggest that while control interventions may substantially reduce transmission in the short term, they may also alter population-level immunity profiles in ways that influence long-term disease persistence and rebound dynamics. The study highlights the importance of incorporating immunity-mediated effects into malaria control planning and provides insights relevant to sustainable intervention strategies in malaria-endemic regions of Sub-Saharan Africa.

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### Predictive modelling of cancer stem cell dynamics in tumour recurrence using ordinary differential equations

**Kaletsane Mukhethwa Chantel** (*University of Venda*)

Tumor recurrence remains a major challenge in cancer treatment, often driven by complex interactions between cancer stem cells (CSCs), the immune system, and the tumor microenvironment. Despite advances in therapy, the mechanisms underlying tumor persistence and regrowth are not fully understood. The aim of this study is to investigate the role of cancer stem cells in sustaining tumor recurrence and to identify key parameters influencing tumor progression within a unified modeling framework. A system of ordinary differential equations (ODEs) incorporating cancer stem cells, non-stem cancer cells, immune cells, and the tumor microenvironment is developed and analyzed. The model is studied using equilibrium and stability analysis, and numerical simulations are performed using a Nonstandard Finite Difference (NSFD) method to preserve essential qualitative properties of the system. Sensitivity and simulation results reveal that cancer stem cell proliferation, immune response parameters, and microenvironmental interactions are key drivers of tumor dynamics. In particular, increased immune recruitment and immune killing rates contribute to tumor suppression, while higher cancer stem cell growth rates and improved microenvironmental conditions promote tumor persistence. The findings demonstrate that cancer stem cells play a central role in sustaining long-term tumor presence and highlight the importance of targeting CSCs for effective tumor control and prevention of recurrence.
**Keywords:** Cancer stem cells, Tumor recurrence, Mathematical modeling, Immune response, Tumor microenvironment, Nonstandard finite difference method

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### Enhanced solution for the advection–diffusion–reaction equation using the Physics-Informed Neural Network technique

**Lekaba Thabo Maomela** (*University of Venda*)

This study focuses on the use of Physics-Informed Neural Networks (PINNs) to solve the 1D Advection–Diffusion–Reaction (ADR) equation. The performance of the PINN model is evaluated against the classical Crank–Nicolson Finite Difference Method (CNFDM) and validated against analytical solutions to assess improvements in accuracy, robustness, and flexibility. Quantitative analysis reveals that the PINN achieved a high level of accuracy, with absolute errors ranging from approximately $2.13 \times 10^{-4}$ to $1.17 \times 10^{-3}$ across the spatial domain. The study uses a neural network architecture with two hidden layers of 80 neurons each, trained in a two-stage process using the Adam and L-BFGS optimizers. This work contributes to the growing field of physics-informed machine learning by demonstrating the strengths and quantitative reliability of the PINN technique for solving complex partial differential equations in transport phenomena.
**Keywords:** Physics-informed neural networks (PINNs), loss function, machine learning, Crank–Nicolson finite difference method (CNFDM)

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### On the numerical approach of MHD mixed convection flow and radiative heat transfer of Sisko nanofluid over nonlinear stretching sheet with viscous Ohmic dissipation

**Nefale Phindulo** (*University of Venda*)

This study numerically investigates the magnetohydrodynamic (MHD) mixed convection and radiative heat transfer characteristics of a Sisko nanofluid over a nonlinear stretching sheet, accounting for viscous and Ohmic dissipation effects. The governing nonlinear partial differential equations describing the flow and heat transfer phenomena were formulated based on the fundamental principles of fluid dynamics and heat transfer. Similarity transformation was employed to convert the governing equations into a

system of dimensionless ordinary differential equations, from which the relevant physical parameters were identified. The resulting nonlinear system was solved numerically using the Spectral Relaxation Method (SRM) and the Spectral Quasi-Linearization Method (SQLM), and the accuracy of the solutions was validated using the MATLAB built-in solver bvp4c. The effects of various governing parameters on the velocity, temperature, and concentration profiles, as well as on engineering quantities of practical interest, were analyzed and illustrated graphically and in tabular form. The numerical findings are compared with existing results available in the literature to establish the reliability and accuracy of the proposed methods. The outcomes of this study show that when the Sisko nanofluid memory increases under high chemical reactions, it makes the nanofluid valuable in industrial settings under the production of lubricants and also in health settings under hyperthermia cancer treatments. It was also observed that both the SRM and SQLM methods converge very quickly in a few iterations.
**Keywords:** MHD, Sisko nanofluid, Viscous Ohmic dissipation, Radiative, convective flow.

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## A comparative study of trivariate spectral collocation and overlapping grids multidomain spectral-linear partition approaches

**Sibonelo Nzama** (*University of KwaZulu-Natal and North-West University*)

This paper introduces the overlapping grids multidomain spectral linear partition method (OGMSLPM), a novel numerical scheme for solving trivariate nonlinear parabolic partial differential equations (PDEs) with Dirichlet boundary conditions. The scheme uniquely combines the linear partition linearisation (LPL) technique with an overlapping grids multidomain approach applied to the time variable, within a spectral collocation framework using Lagrange interpolation. The method is applied to both single equations and coupled systems, and its accuracy, reliability, and efficiency are rigorously benchmarked against the well-established trivariate spectral collocation method (TSCM) by solving four distinct problems with known exact solutions. Numerical results demonstrate that the OGMSLPM consistently improves computational efficiency and numerical stability, particularly for simulations requiring fine discretisations. A key finding is that the condition number of the OGMSLPM system grows at a significantly lower rate, in some cases by an order of magnitude compared to the TSCM, which leads to more reliable solutions at high resolutions. This enhanced performance is attributed to the dual innovation in the method's architecture: firstly, the overlapping time domain strategy, which mitigates the numerical ill-conditioning and restrictive stability limits typically associated with the temporal integration of stiff operators; and secondly, the robust LPL, which requires no derivatives of the nonlinear operator during iteration, ensuring superior numerical stability. Consequently, the OGMSLPM is presented as an efficient and robust alternative to current methods for solving complex, nonlinear PDEs.

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## Nonstandard finite difference schemes for differential equation in combustion theory

**Shamone Ruiters** (*University of the Witwatersrand*), Yusuf Olatunji Tijani

Thermal explosion theory examines how self-heating in exothermic reacting systems drives the transition from a stable state to thermal runaway, leading to ignition or explosion. The mathematical models governing these phenomena are nonlinear differential equations characterized by stiff exponential terms. Standard finite difference methods (SFDM) are often unreliable for such systems, as they produce numerical instabilities and inconsistent results unless restrictively small step sizes are used. To address these computational limitations, this study investigates the construction of nonstandard finite difference (NSFD) schemes for combustion models. The objective is to develop a more robust numerical framework that maintains stability and physical consistency without the step-size constraints typical of conventional methods.

## Entropy analysis of cross trihybrid nanofluid flow with nanolayer thickness using the overlapping grid method

**Samriddha Deb** (*North-Eastern Hill University*)

A non-Newtonian cross trihybrid nanofluid made up of $MoS_2$, Ag, and $TiO_2$ nanoparticles suspended in a CMC-water base fluid is investigated numerically in the current problem. The effects of the nanoparticles' interfacial nanolayer are taken into consideration when analyzing various aspects of the nanofluid flow. The numerical processing is then done using the spectral local linearization technique enhanced with an overlapping-grid scheme. Strong agreement between the current results and previously published results is shown in the comparison table. According to the findings, the velocity profile rises with increase in Weissenberg number. The analysis reveals that the temperature profile is positively impacted by the magnetic parameter, curvature parameter, Biot number, temperature ratio, radiation parameter, and Brinkman number. The Nusselt number, Sherwood number, and coefficient of skin friction are examined. A comparison of the Nusselt numbers for trihybrid, hybrid, and mono nanofluids is examined, emphasizing how higher number of nanoparticles increases the Nusselt number. Nusselt number is examined in relation to interfacial nanolayer thickness as well. Several tables and graphs are used to illustrate and explain the findings.

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## Block hybrid methods for fourth-order boundary value problems

**Mpho Mendy Nefale** (*Cape Peninsula University of Technology and University of the Witwatersrand*)

This study introduces a block hybrid method (BHM) for obtaining approximate solutions to fourth-order nonlinear boundary value problems (BVPs). The approach involves partitioning the domain into discrete blocks, where a set of collocation points is introduced within each block to approximate the unknown solution. By integrating the collocation equations across these points, the BVP is reformulated into a structured system of algebraic equations. The performance of the BHM is validated and compared against spectral methods on single-domain and overlapping grids. The results obtained from the preliminary analysis indicate that the BHM achieves higher accuracy and enhanced numerical stability. Specifically, it is apparent from the results that the BHM maintains a significantly lower matrix condition number compared to established spectral techniques, which often suffer from ill-conditioning as the number of grid points increases. This study confirms that the BHM provides a highly reliable alternative for solving complex fourth-order BVPs.

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## An AI-Driven decision support system for intelligent selection and solution of differential equation models in scientific computing

**Ntandazo Mpisi** (*University of KwaZulu-Natal*)

This project proposes the development of an AI-driven decision support system for the intelligent selection and solution of differential equation models arising in scientific and engineering applications. The system will integrate artificial intelligence, symbolic computation, and advanced numerical methods to automate the process of identifying, classifying, and solving mathematical models, particularly ordinary and partial differential equations. The proposed framework will employ machine learning techniques to classify differential equations based on their structural properties and recommend optimal solution strategies. Analytical approaches will be prioritised where feasible, while robust numerical techniques such as spectral methods, finite difference methods, and hybrid block methods will be utilised for complex or nonlinear models. The system will further incorporate adaptive learning capabilities to

improve decision accuracy over time. In alignment with Fourth Industrial Revolution (4IR) priorities, this research leverages artificial intelligence, computational modelling, and scientific software development to create an intelligent digital tool for advanced problem solving. The system will be implemented as an interactive platform capable of processing user defined mathematical models and generating efficient, interpretable solutions. From a South African perspective, this project contributes to capacity building in high demand digital and computational skills, including artificial intelligence, data science, and numerical modelling. It further supports innovation in scientific computing and engineering analysis, with potential applications in fluid dynamics, energy systems, and advanced materials research.

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## An overlapping multi-domain bivariate simple iteration method for nonlinear partial differential equations

**Mulweli Mudau** (*University of the Witwatersrand*)

Nonlinear partial differential equations (PDEs) arising in fluid dynamics are often stiff, parameter sensitive, and defined on semi-infinite domains. These characteristics make analytical solutions difficult or impossible to obtain, necessitating robust numerical techniques. Moreover, numerical simulations often require extensive computational time intervals to capture the physical dynamics. However, traditional time-stepping schemes may encounter difficulties when the temporal domain is extended to these large intervals. This study presents an overlapping multi-domain bivariate simple iteration method (OMD-BSIM) for solving systems of nonlinear PDEs. The classical bivariate simple iteration method (BSIM) employs relaxation rules to linearize nonlinear partial differential equations (PDEs). Subsequently, the resulting decoupled equations are discretized in both spatial and temporal domains using the Chebyshev pseudospectral method. The proposed OMD-BSIM extends this framework by decomposing the temporal domain into overlapping subdomains, with each new subdomain initialized using an interior point of the preceding subdomain. The OMD-BSIM is applied to a model describing mixed convective flow of an Eyring–Powell fluid subjected to magnetohydrodynamic (MHD) effects and nonlinear thermal radiation over an oscillatory stretching sheet. The proposed OMD-BSIM schemes are compared with the classical bivariate simple iteration method (BSIM) in terms of convergence speed, stability, accuracy, and computational efficiency. These performance metrics are evaluated using condition numbers, residual norms, convergence errors, and the number of discretization points required for convergence. Results show that OMD-BSIM exhibits better computational efficiency, stability, and faster convergence and attains higher accuracy than BSIM. The findings establish that OMD-BSIM provides a robust and efficient framework for solving stiff, nonlinear PDEs on semi-infinite domains, outperforming classical BSIM, and offers a numerical tool for modeling complex fluid-flow systems in applications.

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## Nonlinear interaction of large-amplitude plasma solitary waves in the small amplitude-difference regime

**Daniel Mahlangu** (*North-West University*)

This work is devoted to the numerical investigation of overtaking collisions of large-amplitude, negative-polarity, ion-acoustic solitons. It considers a fluid plasma model with cold ions and Cairns-distributed electrons. The study employs a custom-developed numerical solver that implements the Sagdeev pseudopotential technique to generate the initial conditions. To solve the full system of partial differential equations, the solver uses the method of lines, combining a finite-difference method for spatial discretisation, Newton's method for boundary value problems, and the fourth-order Runge–Kutta method for time integration. This framework is validated against analytical solutions derived via the Hirota bilinear method for the KdV equation, obtained through reductive perturbation theory. The study shows that the elasticity of collisions depends not on the absolute amplitude of the individual

solitons but rather on the relative amplitude between the colliding solitons. Additionally, the study proposes a threshold of relative amplitude below which collisions remain elastic regardless of the absolute amplitude

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### Optimized overlapping and non-overlapping block hybrid methods for nonlinear second-order boundary value problems

**Zwelithini Mkhonta** (*University of Eswatini*)

This paper presents optimized non-overlapping and overlapping block hybrid methods for the direct numerical solution of nonlinear second-order boundary value problems. The methods are derived using an interpolation–collocation framework with optimized collocation points, while quasilinearization is employed to linearize the resulting nonlinear algebraic systems. Theoretical properties of the proposed formulations are investigated through analyses of the local truncation error, consistency, zero-stability, and convergence. The overlapping formulation incorporates information from neighbouring computational blocks through a borrowed collocation point, leading to enhanced inter-block information exchange and improved local approximation properties. The performance of the methods is assessed using several nonlinear benchmark problems, including a nonlinear heat-transfer model, the classical Troesch problem, a coupled nonlinear two-equation system, and a nonlinear three-equation system. Numerical results are presented in terms of maximum errors, pointwise error distributions, convergence histories, and computational cost. The results demonstrate that both methods are accurate and robust, while the overlapping block hybrid method generally provides improved accuracy and enhanced performance for challenging nonlinear boundary value problems. The proposed framework therefore offers an effective approach for the direct numerical solution of nonlinear second-order boundary value problems.
**Keywords:** Boundary value problems, Block hybrid methods, Overlapping block methods, Quasilinearization method, Collocation methods, Nonlinear ordinary differential equations

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### Sensitivity analysis for nanofluid flow driven by thermal and mass diffusion through a rotating convergent/divergent channel

**David Moseki** (*North-West University*)

This talk presents local sensitivity analysis of the nanofluid flow driven by thermal and mass diffusion through divergent/convergent channel. The mathematical model under investigation is formulated using continuity, momentum conservation laws, energy, and concentration equations, along with physical laws representing forces that affects the fluid flow field. The model is solved using numerical spectral quasi-linearisation method and the results obtained are presented graphically to examine the effects of parameters on the fluid flow field. Thus, we perform sensitivity analysis to identify the dominant parameters influencing the fluid flow in the system. The outcomes of this research are expected to provide valuable insights for the design and optimisation of advanced heat and mass transfer systems. For example, applications such as rocket nuzzle design processes involving rotating components.
**Keywords:** Nanofluid flow, local sensitivity analysis, spectral quasi-linearisation method, rotating convergent/divergent channel

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## Bioconvection and magnetohydrodynamic flow of trihybrid nanofluids containing motile microorganisms through porous media: A modified Buongiorno model approach

**Abhijit Sharma** (*North-Eastern Hill University*), Salma Ahmedai, Sabyasachi Mondal, Precious Sibanda

This study investigates the complex heat and mass transfer characteristics of a magnetized trihybrid nanofluid flow containing motile microorganisms over a porous medium. The working fluid consists of a water-based suspension of Cobalt Iron Oxide ($CoFe_2O_4$), Titanium Oxide ($TiO_2$), and Aluminum Oxide ($Al_2O_3$) nanoparticles. By integrating the Tiwari-Das framework with a modified Buongiorno model, the analysis accounts for the critical effects of Brownian motion and thermophoretic diffusion within a ternary nanoparticle system. The governing nonlinear partial differential equations are transformed into a system of coupled ordinary differential equations via appropriate similarity transformations and subsequently solved using a robust numerical scheme. A comprehensive parametric study evaluates the influence of the magnetic field, bioconvection Schmidt number, and porosity parameters on transport profiles. The results demonstrate that the trihybrid configuration significantly enhances thermal performance compared to traditional hybrid and mono-nanofluids. Furthermore, the interaction between the magnetic field and bioconvection parameters is shown to exert a stabilizing effect on flow dynamics, offering critical insights for the optimization of micro-scale bio-thermal systems.
**Keywords:** Bioconvection, motile microorganisms, modified Buongiorno model, Spectral-quasi linearization (SQLM)

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## A meshless RBF-FD solver for compressible Euler's equation with shock-sensor based artificial viscosity

**Gabriel Simaneka Mbokoma** (*University of South Africa*)

The compressible Euler equations are a key model in fluid dynamics that describes compressible, inviscid flows. Recently, meshless numerical methods have emerged as powerful alternatives to traditional mesh-based discretization methods, especially for complex geometries and dynamically changing solution structures. We introduce a meshless Radial Basis Function–Finite Difference (RBF-FD) solver for nonlinear hyperbolic conservation laws based on the compressible Euler equations. In the Medusa C++ library, the solver is designed to be flexible for node creation, stencil fabrication, and differentiation operator assembly. To ensure well-structured, high-order accuracy, spatial discretization is performed using a polyharmonic spline (PHS) radial basis function with polynomial terms. Integration is a third-order strong-stability-preserving Runge–Kutta (SSP-RK3) protocol. The shock-sensor-based artificial viscosity (SSAV) method was integrated to suppress numerical oscillations near discontinuities. The approach is validated in one-dimensional Riemann problems such as the Sod shock tube and the smooth Euler test, and is modified for two-dimensional problems such as shock–vortex interaction. Results showed that the generated methods accurately predict shocks, realistic flow features, and overall stability.

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## Southern African HIV/AIDS dynamics analysis integrating statistical and mathematical modelling

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HIV is a fatal infectious disease spread mainly by sexual transmission recently; that is highly prevalent in endemic areas of the world. A simple subset of an epidemic population model for HIV-AIDS can be split into the following: Susceptible, HIV-Infected and AIDS-infected, (SIA). We propose and use

a nonlinear mathematical model to analyse the dynamics of HIV transmission in a population. The equilibrium points for the model are computed where the stability of the disease-free equilibrium is governed by the basic reproductive number. PRCC technique is utilized to investigate the sensitivity analysis of the model. Numerical simulations are computed to analyse the qualitative behaviour of the model compartments, using MATLAB ode45 solver. Optimal control analysis is introduced where the optimal problem is formulated using Pontryagin's Maximum Principle. The simulations were carried out using the forward- backward sweep method, allowing us to determine the impact of the control strategies on each of the model compartments, particularly the infectious class. The influence of a single control and the combination of both is investigated. The comparison of the two strategies in eliminating the reproductive number is done by simulation of contour plots. It is found that the use of more than one control is the most optimal when compared to the use of a single control.

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## Mathematical modelling of tumor growth dynamics and metastatic spread: McKendrick-von Foerster approach

**Tshepho Chukudu** (*University of Limpopo*)

Mathematical modelling of tumor growth dynamics plays a crucial role in cancer research, providing insight into tumor progression and aiding clinical decision-making. Various models, including exponential, logistic, and Gompertzian functions, are widely used to describe growth over time. This study explores these mathematical models, highlighting their strengths and limitations in predicting tumor behaviour using the McKendrick-von Foerster equation. Although exponential growth models offer simplicity in early growth phases, logistic and Gompertzian growth models provide more accurate representations of growth saturation and deceleration. By comparing these models, this study aims to improve understanding of tumor growth dynamics and contribute to the development of more effective predictive tools in cancer research and treatment. The findings contribute to advancing predictive mathematical Oncology and guiding data driven therapeutic strategies.
**Keywords:** Mathematical oncology, tumor growth, metastasis, McKendrick-von Foerster equation, Gompertz model, exponential model, and population dynamics

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## Lie symmetry analysis of the Lane-Emden equation

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This study explores the Lane-Emden equation, a classical model in astrophysics that describes the equilibrium structure of self-gravitating spherical gas clouds. To analyze this equation, we employ Lie symmetry analysis, a powerful technique for identifying transformations that leave a differential equation invariant. By determining the symmetries of the equation we reduce the original nonlinear second-order differential equation into simpler, more tractable forms. Our investigation reveals that the symmetry structure of the Lane-Emden equation is closely tied to the nonlinearity parameter $r$. For physically significant values of $r$, specifically $r = 0, 1$, and 5, we derive exact analytical solutions. These solutions include both well-known results and new findings, demonstrating the efficacy of the symmetry method in systematically generating meaningful solutions. Beyond recovering classical results, this approach offers a rigorous and structured framework for uncovering new insights into nonlinear equations. The methods and results presented in this work underscore the utility of Lie symmetry techniques in understanding complex physical models, particularly in the study of stellar structures and other spherical systems in astrophysics.

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